

Promethium

Pm (Pm I). Strongest classified transitions below 2 μm (NIST): 23 levels, 16 lines.

Pm + ionisation limit 45020.80 cm⁻¹ = 5.58187 eV (222.120 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in $e a_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A . Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and measured lifetime (where one exists).

